

filling, and an algorithm is used to redistribute the energy to more efficient subcarriers to support higher data rates. The number of allocated bits per subcarrier is then rounded to the nearest integer. This is known as rate-adaptive bit-loading.

American National Standard Institute (ANSI) standard T1.413 (1997) provides a frequency range of 0 to 26kHz for POTS signal. The frequency range from 30Hz-4000Hz is used for voice, and the range from 4Hz-25kHz is unused guard band. 26kHz-1130kHz frequency band accommodates 256 sub-channels, each 4.3125kHz wide. Centres of these sub-channels are also kept separated by 4.3125kHz. The centre frequency of each sub-channel is $N \times 4.3125\text{kHz}$. The spectrum of each sub-channel overlaps that of its adjacent. It is not confined to a 4.3125kHz-wide channel.

The orthogonality of DMT makes this possible without interference. Sometimes, a small guard band is also used between upstream and downstream channels in order to prevent interference. The frequency layout of ADSL signal using FDD is shown in Fig. 3.

Individual carriers in the upstream and downstream ranges are quadrature amplitude modulated to carry two bits per second per hertz to 15 bits per second per hertz. The bit-carrying rate of each band is allocated adoptively (rate-adaptive bit-loading) during the initialisation process, that is, the bit-carrying rate is decided by the SNR in the relevant channel in accordance with equation (3). The larger the SNR, the larger the signal space allocated to the band. If the SNR of a band is too small, that band may not be utilised. Sub-channel energy is redistributed to more efficient sub-channels that ultimately yield higher data rates.

Theoretically, it has been found from Shannon's theorem that the maximum information transmitted using quadrature amplitude modulation (QAM) is 64.7kbps/channel. So, the maximum transfer rate for 256 chan-

nels results in 16Mbps. But, the circuit design constraints and attenuation in copper lines at high frequencies restrict high data-transfer rate. The attenuation in a conductor is dependent on the frequency of operation and is given by:

$$\alpha = \frac{P_t}{2P_T} \dots\dots\dots(5)$$

where, P_T is power transmitted that is decided by Poynting theorem and is given by integrating the real part of complex Poynting vector, and P_t is power loss.

At high frequencies, the current is confined to the surface of the copper line. The depth of penetration of electromagnetic wave, known as skin depth (d), is given by:

$$d = \sqrt{\frac{R_s}{\pi f \mu_0 \mu_r}} \dots\dots\dots(6)$$

where, f is the frequency of operation, R_s is resistivity of the conductor wire, μ_0 is absolute permeability and μ_r is relative permeability of the conductor wire material.

Hence, it is clear that there exists only surface current on the surface of the conductor.

Now, for metallic conductors of high conductivity, like copper pair lines, Ampere's circuit law can be used to calculate surface current density J_s and it is expressed as:

$J_s = n_t \times H$
where, n_t is the normal to the surface of the conductor. Then, the time-averaged power-loss per square metre due to current flow in conducting surface is given by:

$$P_L = \frac{1}{2} |J_s|^2 R_s \dots\dots\dots(7)$$

Hence, time-averaged power-loss per unit length of the copper line is given by:

$$P_L = \frac{1}{2} \int_{\text{surface area}} |J_s|^2 R_s ds \dots\dots\dots(8)$$

where, the integration is taken over the wall-surface of a unit length of the copper line, and R_s is the resistance of the conductor for a unit length and unit width, and is called surface resistivity.

So, with all these constraints, only 8Mbps data-transfer rate is feasible, which is half the capacity.

In echo-cancelling duplex (ECD), another method of allocating channels for upstream and downstream bands in ADSL, frequencies allocated for upstream data flow are used for both upstream and downstream data flow. This leads to a higher capacity for downstream flow since the lower 112kHz of the ADSL range contains better channels because at higher frequencies attenuation rises.

In echo-compensation technique, the same frequency channel at the same time is used to send and receive information. The last transmitted data is preserved by the transceiver. The data received by the transceiver contains both the received data as well as the transmitted data (which is actually an echo). The transceiver then uses the retained transmitted information to cancel out the echo from the received signal. The transmitted and received information is separated using highly-precise echo-cancellation algorithms and high-speed digital-signal processors.

Conclusion

The Internet is used largely for downloading files, HTML and graphical content, while uploading data files is rare and limited to very few users. This type of bandwidth requirement can be efficiently met by ADSL technology.

ADSL caters specifically to connections between ISPs and subscribers, and uses FDD to superimpose the upstream and downstream data over POTS signal onto existing copper lines. A splitter at the customers' premises separates the POTS signal from the high-frequency upstream and downstream data. These high-frequency data are routed through the modem to subscribers' computers. The information-carrying capacity and robustness of ADSL wire-line broadband technology make it a successful high-speed Internet access technology. ●